Identifying Structures in the Heart

Exam code: 0610 & 0970

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Written by: Lára Marie McIvor
Reviewed by: Lucy Kirkham
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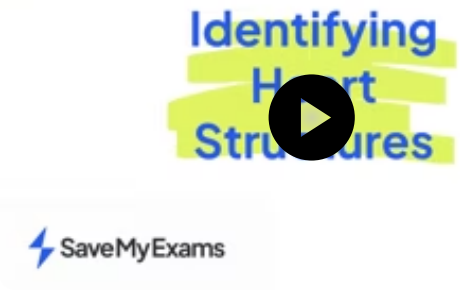

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Identifying Structures in the Heart: Extended

Extended Tier Only

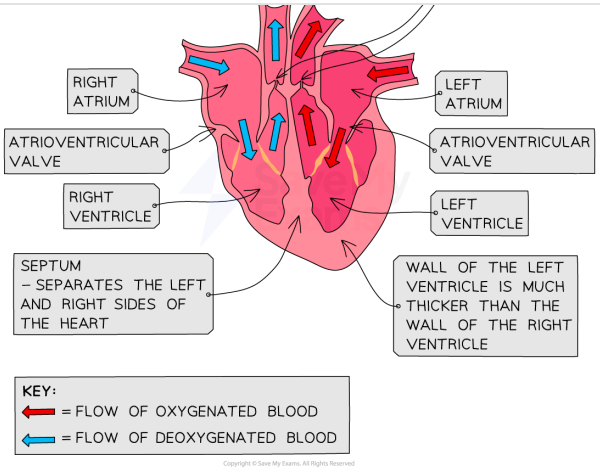
- The **ventricles** have thicker muscle walls than the atria as they are pumping blood out of the heart and so need to generate **a higher pressure**
- **The left ventricle has a thicker muscle wall than the right ventricle** as it has to pump blood at high pressure around the **entire body**, whereas the right ventricle is pumping blood at lower pressure to the **lungs**
- The **septum** separates the two sides of the heart and so **prevents mixing of oxygenated and deoxygenated blood**

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



The structures in the heart maximise the efficiency with which it can pump blood around the body

The function of valves

- The basic function of all valves is to **prevent blood from flowing backwards**
- There are two sets of valves in the heart:
  - The **atrioventricular valves** separate the atria from the ventricles
  - The valve on the right side of the heart is called the **TRICUSPID** and the valve on the left side is called the **BICUSPID**
  - These valves are pushed **open when the atria contract** but when the **ventricles contract they are pushed shut** to prevent blood from flowing back into the atria
  - The **semilunar valves** are found in the two blood arteries that come out of the top of the heart
  - They are unusual in that they are the **only two arteries in the body that contain valves**
  - These valves **open when the ventricles contract** so blood squeezes past them out of the heart, but then shut to avoid blood flowing back into the heart

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